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Adapting the Biomek® 2000 Laboratory Automation Workstation for Printing DNA Microarrays

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ABSTRACT

The Biomek® 2000 Laboratory Automation Workstation is used for liquid handling and other repetitive operations in many laboratories. Since it has very good spatial positioning capabilities, we have modified this workstation to deliver samples at high densities onto microscope slides to produce DNA microarrays. The workstation tool, originally designed for bacterial colony replication, was adapted to carry special printing pins and was further modified to improve its positional accuracy. Software written in the Tool Command Language was concurrently developed to control the movements of the workstation arm during the process of printing. With these modifications, the workstation can reliably deliver individual samples at a spacing of 0.5 mm, corresponding to a total of more than 3000 samples on a single slide. Arrays prepared in this way were successfully tested in hybridization experiments.

INTRODUCTION

The recently developed DNA microarray technology (1,5) represents a further step towards high-throughput analysis of complex biological samples. It involves thousands of DNA samples (complete cDNAs or expressed sequence tags [ESTs]) immobilized at very high density on solid supports, usually glass surfaces. Such "DNA chips" are then hybridized with fluorescently labeled probes derived from mRNAs isolated from tissues of interest, and the arrays are scanned to quantitatively evaluate hybridization signals produced from the individual dots. This arrangement allows parallel monitoring of all transcripts with corresponding cDNAs that have been placed on the chip. Moreover, direct comparison of gene expression in two tissues can be made by simultaneous hybridization of corresponding probes labeled with different fluorochromes. This approach has been successfully used to study gene expression in *Arabidopsis* (6), human (2,7) and yeast (3).

In spite of the advantages offered, the widespread use of DNA microarrays is currently restricted by difficulties associated with equipment allowing the high-precision delivery of DNA samples. A custom-made instrument designed especially for this purpose has been described (8); however, there are technical difficulties associated with its construction, and it represents a substantial financial investment.

The Biomek® 2000 Laboratory Automation Workstation (Beckman Instruments, Fullerton, CA, USA) is used for liquid handling and other automated operations in many laboratories. Since it has very good spatial positioning capabilities, we have investigated whether it might be possible to modify this workstation to deliver samples at high densities onto microscope slides to produce DNA microarrays. In this report, we describe low-cost modifications that allow reproducible printing of arrays on glass containing up to 3000 DNA elements.

MATERIALS AND METHODS

Configuration of the Workstation

The Biomek 2000 Workstation was equipped with right- and left-side modules (Part Nos. 609047 and 609048; Beckman Instruments) and installed in a Nuaire™ Class II Type A/B3 Laminar Flow Hood (Nuaire, Plymouth, MN, USA). Most of the components used for DNA printing [e.g., the high density replicating (HDR) tool used as the basis for constructing the DNA printing tool, the air fan and three reservoirs] were purchased as a part of the 96 HDR System (Beckman Instruments). Four labware holders (Part No. 609120; Beckman Instruments) were used for positioning the microplates and reservoirs on the Biomek 2000 work surface. The individual components were located as follows (see also Figure 2b): Position A1 - fan; Position A2 - reservoir filled with 96% ethanol; Position A3 - microplate containing DNA samples; Position B1 - reservoir filled with deionized water; Position B2 - microplate covered with three sheets of Scott® Absorbent Wipes. The samples were spotted directly from Concord PCR microplates (MJ Research, Watertown,

MA, USA) that were positioned on the work surface within a deep-well plate (Part No. 267006; Beckman Instruments). A glass plate (310 × 260 × 5 mm) was aligned with the right side of the work surface and anchored to the work surface in this position using Scotch™ Double-Coated Tape. The plate served as a support for up to 28 microscope slides that were attached to its upper surface using double-sided tape.

Design of the Printing Tool and Pins

The printing pins were machined from 9.5-mm diameter 304 stainless steel rod. The rod was cut into 35-mm sections. Figure 1 shows the steps in the machining process. The rod was first turned to a diameter of 3.15 mm to a distance of 19 mm (Figure 1b). Next, this end was machined to a conical tip subtending a 10° angle (Figure 1c). The other end of the rod was then turned to a diameter of 3.15 mm to a distance of 14.3 mm. This leaves a 9.5-mm-diameter collar having a width of 1.7 mm (Figure 1d). Finally, a slit was cut in the conical end of the rod to the depth corresponding to base of the cone, using a 0.25-mm slit saw (Figure 1e). Before use, the pins were cleaned by sonication in 0.1% sodium dodecyl sulfate (SDS) for 1 h and washed in water and 96% ethanol.

The tips were placed in a specially constructed assembly with the dimensions of the HDR tool. Two plates were fabricated from 3.2-mm-thick stainless steel. These were spaced apart using two 25.4- × 12.7- × 6.4-mm spacers and were held in place by two screws. Four holes (3.2-mm diameter each) were drilled in the center of the plates as a 9- × 9-mm array. Springs taken from retractable ball-point pens were used to seat the printing tips against the lower of the two plates.

Software

The whole procedure of spotting samples and washing the printing pin is controlled by a program written in Tool Command Language (Tcl) (4), which can be launched directly from the workstation control environment (BioWorks™ 2.0; Beckman Instruments). The program script and detailed instructions about alignment of the instrument prior to printing are available (either from the authors or from their Web page (**http://latin.arizona.edu/~dgalbrai/robot/**).

Determination of Spotting Accuracy

To determine the accuracy of printing, arrays comprising 96 (12 × 8) DNA samples dissolved in 2× standard saline citrate (SSC) were spotted onto coverslips. Dry microarrays were projected at a 20-fold magnification onto a computer-designed print of an "ideal" grid, and the distance of the grid marks from the actual centers of the sample dots were

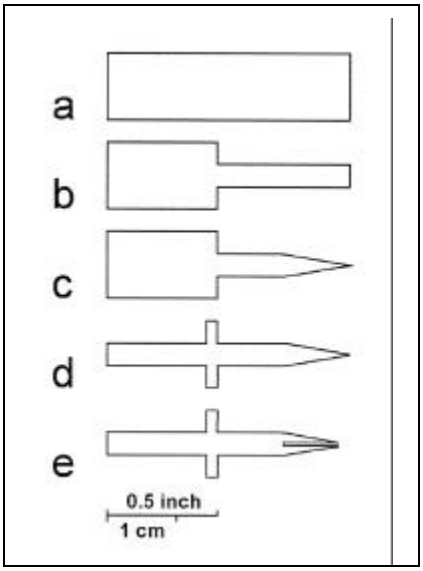


Figure 1. Individual steps in manufacturing printing pins. See text for detailed description.

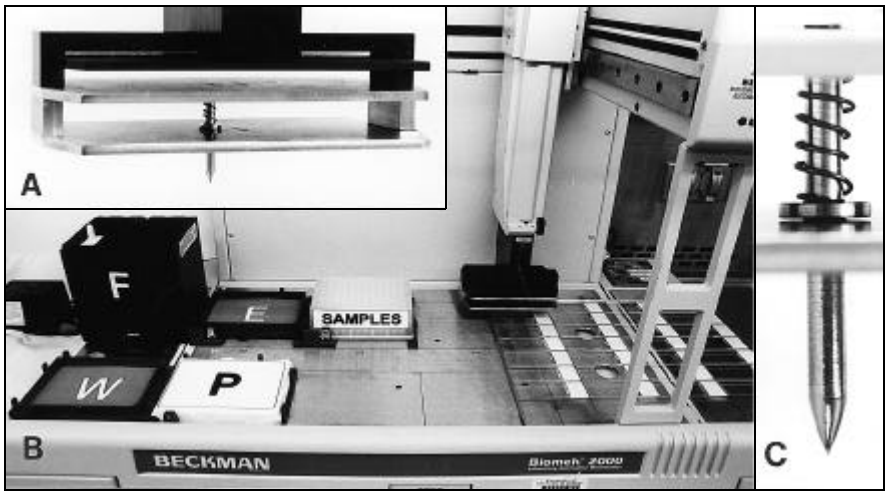


Figure 2. The Biomek 2000 workstation equipped for printing DNA microarrays. (A) Detail of modified printing tool. (B) Layout of individual components on the work surface. The slides are arranged in four columns on the right side of the work surface. Abbreviations: F - fan, E - ethanol reservoir, W - water reservoir and P - paper pad. (C) Detail of the printing pin.

determined. The precision of printing was expressed as standard deviation of positioning along the x- and y-axes.

DNA Samples and Hybridization Conditions

The samples for control hybridizations were prepared by polymerase chain reaction (PCR) amplification using 5′ amino-modified primers and immobilized after spotting on silane-coated slides (CEL Associates, Houston, TX, USA) as described by Schena et al. (7). M13 reverse and T7 primers were used for insert amplification from pBluescript® II SK plasmids (Stratagene, La Jolla, CA, USA). A positive control was made by amplification of a 1120-bp *Eco*RI-*Bam*HI fragment (position 21 227-22 347) of phage λ DNA inserted into this plasmid. The negative control results from amplification of pBluescript II SK lacking inserts, which produces a DNA fragment of 207 bp corresponding to the polylinker region. Hybridization probe was prepared by labeling the λ DNA with fluorescein-dUTP, using the Vistra™ Random Prime Labeling Kit (Amersham Pharmacia Biotech, Piscataway, NJ, USA).

Microarrays were denatured in water for 2 min at 95°C, transferred quickly to 96% ethanol for 20 s and air-dried. Eight microliters of hybridization mixture (5× SSC, 0.1% SDS) containing denatured probe was applied onto slides pre-heated to 65°C, and hybridization was done in a humid chamber at 60°C overnight. The slides were washed in 5× SSC, 0.1% SDS for 5 min; 0.2× SSC, 0.1% SDS for 5 min; and in 0.2× SSC for 2 min, and mounted for observation using an Axiovert® 100TV Confocal Microscope equipped with 2.5× objective (Carl Zeiss, Thornwood, NY, USA).

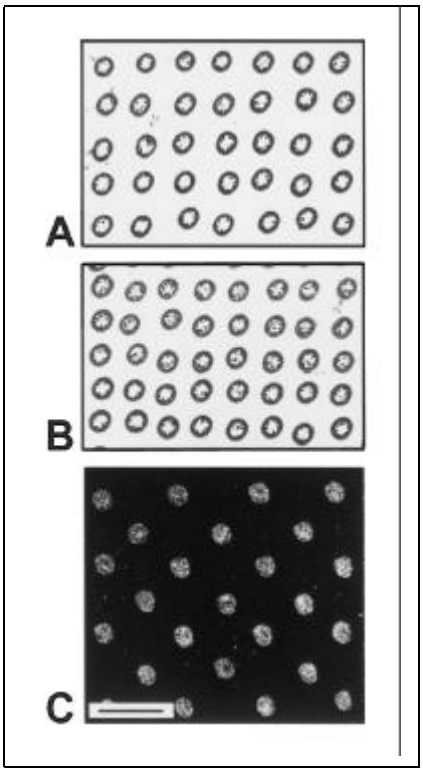


Figure 3. DNA microarrays. DNA samples spotted at 600 (A) and 500 μm (B) element spacings (pictures were taken by projecting the arrays directly onto photographic paper). (C) Hybridization of a fluorescently labeled λ DNA fragment to a microarray in which the elements alternate PCR fragments corresponding to the pBluescript II SK polylinker region containing or lacking the specific λ insert. (Bar = 1 mm).

RESULTS AND DISCUSSION

To adapt the Biomek 2000 workstation for printing DNA microarrays, only minor hardware modifications and the development of proper software were needed. The workstation tool originally designed for bacterial colony replication was adapted to carry special printing pins (Figures 1 and 2). The pins were based on a design described by P. Brown (Reference 8 and Web site **http://cmgm.stanford.edu/pbrown/arrayer.html**), in which the tool body was modified to improve the positional accuracy of the printing pins. This involved stabilizing the pins using two metal plates and small springs (Figure 2a). Up to four printing pins can be used simultaneously, allowing faster processing when high numbers of samples are to be spotted. The applied mechanism of spotting is that described by Shalon et al. (8): about 2 μL of the DNA sample are drawn up into the pin slot by capillary action. Small droplets (ca. 5 nL) are then delivered to the slides when their surfaces are touched by the pin. Since tens of dots can be printed with one load, multiple slides are typically prepared at the same time.

A software program to control the movements of the printing tool was

written in the Tcl language. The program was designed to allow reliable and accurate spotting, and included efficient washing steps to avoid sample cross-contamination on the printing pin. Figure 2b depicts the layout of the individual components used for printing. Except for the glass plate used as a support for slides and the modified printing pins and assembly described previously, all necessary parts are available from Beckman Instruments.

The spotting procedure starts with loading the sample from one well of the microplate. The sample is then spotted in a defined position on up to 28 slides. The remaining sample is then removed from the slot by touching the printing tip against a pad of dry paper. This is followed by sequential washes in water and then ethanol. Most of the ethanol is removed by again moving the tip to the paper pad; residual ethanol is then evaporated by moving the spotting tool to the fan station for 10 s. The tip is then ready for loading and spotting the next sample.

To test the instrument's accuracy, we spotted DNA samples resuspended in 2× SSC at various densities. The resulting arrays were inspected visually to score for "fused" adjacent bands. The error of spotting was also calculated by measuring the deviation of the spots from ideal positions. For spotting to a square grid with nominal x- and y-repeat values of 500 μm, the standard deviation (n = 768) was ± 52.6 μm (x-dimension) and ± 63.7 μm (y-dimension). Evidently, the maximum spotting density depends also on the drop size. For a spot size of approximately 125 μm, reliable arraying was achieved with spacing between the samples down to 500 μm (Figure 3, a and b). Although not as great a density as can be achieved with custom arraying instruments [sample spacing of 380 μm (8)], this allows immobilization of up to 3000 samples upon a single microscope slide, which is sufficient for many research and diagnostic purposes.

The microarrays prepared as described above were tested by hybridization with fluorescein-labeled probes. DNA fragments obtained by PCR amplification of a λ DNA insert or of the pBluescript II SK multiple cloning site alone were spotted in an alternating pattern onto silane-coated microscope slides. Figure 3c shows an example of hybridization of such an array using probe derived from λ DNA, which therefore detects only those array elements containing the λ sequence. A hybridization signal was restricted to every second dot, indicating minimal cross-contamination between different samples. This confirms the effectiveness of the pin washing steps used during the printing of the microarray. The relatively even signal intensity between hybridizing samples indicates that differences in the DNA amounts between replicate dots are minimal.

In conclusion, the results we presented document modification and use of the Biomek® 2000 for printing DNA microarrays of up to 3000 elements per microscope slide. The printing pins and all parts needed for constructing the printing tool can be easily manufactured in a machine shop for approximately $1000. Since the Biomek can also be used for a large variety of liquid transfers, our modifications allow automation of the whole process of making DNA microarrays (handling of bacteria, preparation of PCR products and spotting of DNA) with a single instrument.

REFERENCES

1. **Castellino, A.M.** 1997. When the chips are down. Genome Res. *7*:943-946.
2. **DeRisi, J., L. Penland, P.O. Brown, M.L. Bittner, P.S. Meltzer, M. Ray, Y. Chen, Y.A. Su and J.M. Trent.** 1996. Use of a cDNA microarray to analyse gene expression patterns in human cancer. Nature Genet. *14*:457-460.
3. **DeRisi, J.L., V.R. Iyer and P.O. Brown.** 1997. Exploring the metabolic and genetic control of gene expression on a genomic scale. Science *278*:680-686.
4. **Ousterhout, J.K.** 1994. Tcl and the Tk Toolkit. Addison-Wesley Publishing, Reading, MA.
5. **Schena, M.** 1996. Genome analysis with gene expression microarrays. Bioessays *18*:427-431.
6. **Schena, M., D. Shalon, R.W. Davis and P.O. Brown.** 1995. Quantitative monitoring of gene expression patterns with a complementary DNA microarray. Science *270*:467-470.
7. **Schena, M., D. Shalon, R. Heller, A. Chai, P.O. Brown and R.W. Davis.** 1996. Parallel human genome analysis: microarray-based expression monitoring of 1000 genes. Proc. Natl. Acad. Sci. USA *93*:10 614-10 619.
8. **Shalon, D., S.J. Smith and P.O. Brown.** 1996. A DNA microarray system for analyzing complex DNA samples using two-color fluorescent probe hybridization. Genome Res. *6*:639-645.

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